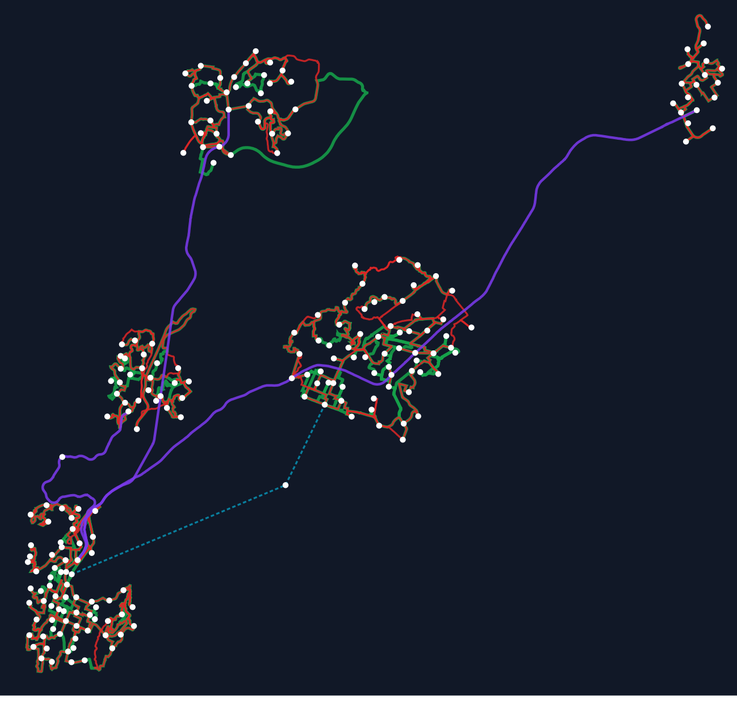
ENGR489 PROJECT · VICTORIA UNIVERSITY OF WELLINGTON

HUNTING MR. X
WELLINGTON EDITION

A new Open Day demo:
a classic board game, played online on a real map of Wellington

Ming Bao · Supervisors: Jens Dietrich, Stuart Marshall

Escooter · Bus · Train · Ferry · Invisible · Double



KEY FINDINGS

Open Day
played live by visitors
on the stand's laptops

216 places
on a real map of Wellington
by bus, train, ferry and e-scooter

No install
join from a web browser
with a link or a six-letter code

WHY OPEN DAY?

- The old software demo was over five years old
- Visitors need something to try in a few minutes
- A game set in their own city starts conversations

HOW TO PLAY

- One player is Mr X; up to five detectives hunt him
- Everyone travels by bus, train, ferry or e-scooter
- Mr X stays hidden, shown only every few rounds
- Catch him within 24 rounds to win

SAME GAME, DIFFERENT SECRETS

Mr. X sees

PLAYERS

Alice	(you)	Node 1
Bob		Node 241
Carol		Node 149

Game server
sends each player only what they are allowed to see

Detective sees

PLAYERS

Alice		?
Bob	(you)	Node 241
Carol		Node 149

He always knows where he is

They see only a “?”

1 **2** 3 4 5 6 7 **8** 9 10 11 12 **13** 14 15 16 17 **18** 19 20 21 22 23 **24**

24 rounds. Mr X is revealed only in the amber rounds; the rest of the time he is a “?”.

TECHNICAL CHALLENGES

KEEPING A SECRET

Each player is sent their own view, so Mr X's position never reaches a detective's screen.

STAYING IN SYNC

Every move is pushed live to every screen. A dropped connection catches up by itself.

A FREE REAL MAP

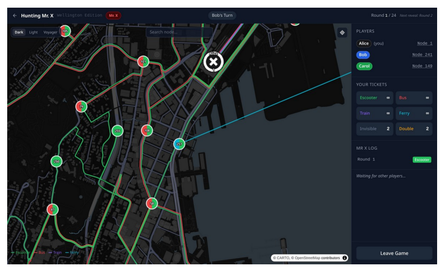
Routes along real roads are worked out once, not paid for on every move.

PROVING IT WORKS

135 automated tests, including 300 random games. Every planted bug was caught.

TECH STACK

Screens	Vue 3 · TypeScript · Tailwind CSS
Map	MapLibre GL · OpenStreetMap
Server	Java 21 · Spring Boot
Live	WebSockets · STOMP
Testing	JUnit 5 · Mockito · PIT
Hosting	Docker · Cloudflare Tunnel



Mr X's screen in a live game

OUTCOME

- Played live at Open Day
- Anyone can join over the internet
- Ready for future Open Days

NEXT STEPS

- A big-screen view for the stand
- A phone-friendly layout
- Computer players for solo games

Source code: scan the QR code.

Board-game mechanics used with permission from Ravensburger, for non-commercial academic use.